

MAHAKUMBH MP HACKATHON

MahaFlow

Adaptive Mobility Intelligence System
for Large Religious Gatherings

Transportation & Mobility Management



Fuzzy Logic AI



Live Network



Route Advisor



Full Transparency

Deployment - <https://mahaflow-theta.vercel.app/>

Mobility at Mahakumbh Scale

 **400M+**

Pilgrims over 45 Days

 **30+**

Interconnected Transit Corridors

 **0**

Margin for Error

The Gap — Traditional dashboards show raw data:

"Traffic is red." → But authorities need actionable intelligence:

"Why is it red — and what should we do right now?"

Intelligence, Not Just Analytics

✓ MahaFlow IS

- ✓ A Fuzzy Logic Reasoning Engine
- ✓ Explainable, auditable decisions
- ✓ Transport-aware smart routing
- ✓ An operational command center

✗ MahaFlow is NOT

- ⚡ An LLM wrapper
- ⚡ A black-box prediction model
- ⚡ A generic maps application
- ⚡ A generic SaaS dashboard

Public-safety AI must be deterministic. MahaFlow's fuzzy logic engine runs offline, processes in milliseconds, and produces 100% reliable outputs every time.

ARCHITECTURE

How MahaFlow Thinks & Reacts



KEY SYSTEM METRICS



Every 5 seconds:

The simulation engine generates a new tick of data → fed into the Fuzzy Logic Engine → 16 parallel rules evaluated → risk score 0-100 for every route → Dijkstra's pathfinding updates the UI instantly.

Configurable Fuzzy Reasoning

5 DYNAMIC INPUT VARIABLES

1. Crowd Density

2. Vehicle Density

3. Parking Occupancy

4. Event Intensity

5. Weather Impact

STEP 1 - Fuzzification

Trapezoidal membership functions classify inputs as LOW, MEDIUM, or HIGH — allowing partial membership (e.g. 60% Medium, 40% High).

STEP 2 - Rule Evaluation

Evaluates 16 concurrent rules using AND-semantics (minimum activation). Rules are plain data — no code changes needed to tune them.

STEP 3 - Defuzzification

Calculates a weighted average across all activated rules to produce a final 0-100 risk score per route.

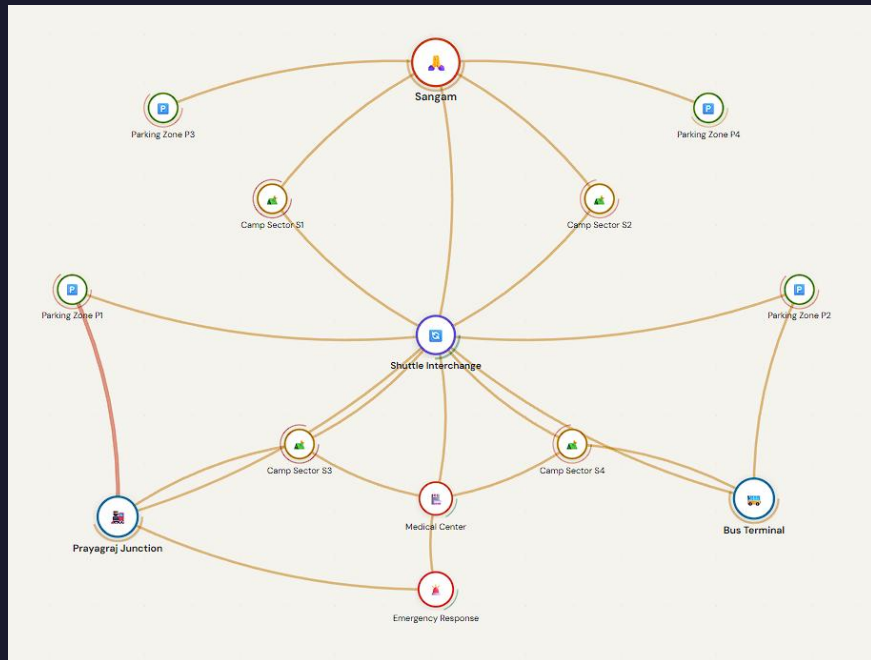
Every Decision Is Transparent

EXAMPLE OUTPUT — CRITICAL ROUTE ANALYSIS

Final Risk Score	78 / 100
Status	⚠️ CRITICAL
Triggered Rule	R5: High crowd + High vehicle → Critical (92% activation)
Crowd Density	HIGH (0.88 membership)
Vehicle Density	HIGH (0.76 membership)
Recommendation	"Divert traffic via alternate corridor. Deploy crowd marshals at Junction B."

 If the AI tells a commander to block a major artery, that commander needs to know exactly WHY. MahaFlow always shows its work.

The Live Transportation Network



LOW risk MEDIUM risk CRITICAL (pulsing)



Topological Layout

Subway-style clarity over geographic noise. Built for operators, not tourists.



Dynamic Styling

Route thickness scales with risk (1.8px → 6px). Critical routes pulse.



Color Coding

Green → Amber → Red → Dark Red. Instant at-a-glance network health.



Interactive

Click any node or route for instant deep-dive fuzzy logic analysis.

Transport-Aware Route Advisor

5 Transport Modes with Distinct Weightings



Walking

Heavy penalty on crowd density



Shuttle Bus

Balances crowd & vehicle traffic



Bus

High penalty for vehicle density



Private Vehicle

Standard traffic weighting



Emergency

Extreme priority — overrides all

Algorithm: Dijkstra's with mode-specific dynamic edge weights

Output: Primary Route + Safest Alternative — so pilgrims always have a backup option.

Authority Command Dashboard

01



Critical Routes

Live sorted list of the most dangerous corridors, updated every 5 sec.

02



Parking Status

Real-time utilization bars for P1-P4, predicting pedestrian surge 30 min ahead.

03



Zone Health Grid

Color-coded grid for instant node status at a glance across the network.

04



Risk Distribution

Macro view: count of routes per risk band (Low → Critical).

05



Operational Recommendations

Direct, actionable advice auto-generated by the fuzzy engine.

06



Fuzzy Logic Analysis

Deep-dive panel: full reasoning chain for the highest-risk corridor.

Continuous System Intelligence

LIVE SIMULATION ENGINE

- ⚡ Sinusoidal crowd curves model morning arrivals, ritual peaks, evening ebbs
- ⚡ Event surges simulate Shahi Snan mass movements
- ⚡ 5-second ticks = 10 simulated minutes of event time
- ⚡ Realistic parking saturation and route pressure propagation

TRILINGUAL SUPPORT

English

Risk: CRITICAL

हिंदी

जोखिम: अत्यधिक

मराठी

धोका: अत्यंत

Zero load time – instant switch

All alerts, rule descriptions, and operational recommendations are fully localized – supporting every operator regardless of language.

MahaFlow

Adaptive Mobility
Intelligence System

MAHAKUMBH MP HACKATHON

Thank you.

READY FOR DEPLOYMENT



Not an LLM Wrapper

Deterministic fuzzy logic — lightning-fast, fully offline.



Total Transparency

Every decision explained. Every rule shown. Full operator trust.



Authority-First Design

Editorial aesthetics that communicate control, not panic.



Zero Infrastructure

Runs in-browser on a ruggedized laptop with spotty internet.



Highly Adaptable

Logic thresholds are data — editable for any global event, no coding.

Omkar Arvind Kottawar